

Pearl's Probability of Actual Causation --- discussion proposal

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Date: 2026-05-26 **Status:** Draft for review — nothing inserted into the paper yet.

The document is organised in two parts. **Sections 1–7** are the conceptual walk-through and motivate the comparison; they're meant to be read top-to-bottom. **Proposal A** and **Proposal B** at the end are paste-ready LaTeX inserts for the paper itself.

1. The construction

Pearl, *Causality* (2nd ed.), Ch. 10 defines the **probability of actual causation** as

$$P(\text{caused}(x, y \mid e)) = \frac{P(U_{xy} \cap U_e)}{P(U_e)} \quad (\text{Def. 10.3.5})$$

where U_{xy} is the set of noise states $\mathbf{u}$ in which the chosen actual-causation predicate “ $X=x$ is an actual cause of $Y=y$ ” holds, and U_e is the set of states compatible with the observed evidence e .

The predicate plugged in is Pearl’s causal-beam definition (1st ed.) or the Halpern–Pearl 2005 AC1–AC3 (2nd-edition postscript and Halpern’s *Actual Causality* book).

The formula is Bayes’ rule applied to actual causation. Each $\mathbf{u}$ determines a deterministic world through the structural equations, so “is $X=x$ an actual cause of $Y=y$?” has a yes/no answer once $\mathbf{u}$ is fixed; U_{xy} collects the yes-states. The ratio rewrites cleanly to

$$P(\text{caused}(x, y \mid e)) = P(\mathbf{u} \in U_{xy} \mid \mathbf{u} \in U_e),$$

i.e. *the posterior probability, given the evidence, that we are in a noise state for which the actual-cause verdict holds*. The two anchoring cases: $U_e \subseteq U_{xy}$ yields probability 1; $U_{xy} \cap U_e = \emptyset$ yields 0.

This is Pearl’s own bridge from “Halpern actual cause” to “*probability of* actual cause” — the same bridge our paper proposes to build with PCI. **A reader of the 2nd edition will notice if we don’t address it.** The rest of this document designs the discussion that places PCI alongside Def. 10.3.5: where the two agree, where they genuinely differ, and why both differences matter.

2. Pearl’s worked example: the desert traveler

The canonical illustration (Pearl Ch. 10, §10.3.3). A traveler has two enemies: enemy 2 shoots and empties the canteen ($X=1$); enemy 1, unaware, poisons the water in the canteen ($P=1$). The traveler dies ($Y=1$). Who is the actual cause of death?

There is one source of uncertainty, packed into a binary noise variable u :

- $u = 0$: the traveler drank the poisoned water before the canteen was emptied — cyanide killed him; the shooter is irrelevant.
- $u = 1$: the canteen was empty before he could drink — dehydration killed him; the poisoning is irrelevant.

Pearl’s beam analysis (or HP05 equivalently) at each state gives the actual-cause verdicts:

- In $u = 1$: $X=1$ is the actual cause; $P=1$ is not.
- In $u = 0$: $P=1$ is the actual cause; $X=1$ is not.

So $U_{x,y} = \{u = 1\}$ and $U_{p,y} = \{u = 0\}$.

Without any extra evidence (just $X=1, P=1, Y=1$), both noise states are compatible, and Def. 10.3.5 reduces to the prior:

$$P(\text{caused}(x, y \mid e)) = P(u = 1 \mid e), \quad P(\text{caused}(p, y \mid e)) = P(u = 0 \mid e).$$

Sharpen the evidence with a forensic report — *no cyanide in the body* — which is incompatible with $u = 0$. Then $U_e = \{u = 1\}$, and

$$P(\text{caused}(x, y \mid e)) = \frac{P(\{u = 1\} \cap \{u = 1\})}{P(\{u = 1\})} = 1.$$

The forensics drives the probability that the shooter was the actual cause to certainty. A toxicology report finding cyanide would do the symmetric thing for the poisoner.

3. PCI on the same example

To put the two methods side by side, we run PCI on Pearl's model.

Model recap. Endogenous $X, P, C, D, Y \in \{0, 1\}$; exogenous binary u ; structural equations from Pearl (p. 323):

$$c = p(u' \vee x'), \quad d = x(u \vee p'), \quad y = c \vee d.$$

Factuals: $X = 1, P = 1, Y = 1$, with prior $P(u = 0) = P(u = 1) = \frac{1}{2}$. The factual values of (C, D) are $(1, 0)$ at $u = 0$ and $(0, 1)$ at $u = 1$.

PCI ingredients. Suspect set $\mathbf{S} = \{X, P\}$, candidate causes the two singletons. Witness pool $\mathbf{W} \subseteq \{C, D\}$ gives four possible active witness sets $\mathbf{T} \in \{\emptyset, \{C\}, \{D\}, \{C, D\}\}$. The alternative-value distribution Δ deterministically flips $1 \rightarrow 0$. The AC-aligned kernel $\Phi(\mathbf{C}, \mathbf{T}) = p^\Gamma \cdot \mathbb{I}[Y^s] \cdot Y^n$ reduces, in this binary deterministic case, to $\Phi > 0$ iff Y flips when $\mathbf{C}$ is set to its alternative and $\mathbf{T}$ is pinned at its factual value.

Necessity check, cause by cause and state by state. Tracing the structural equations:

Cause $\mathbf{C} = \{X\}$, *intervention* $X \leftarrow 0$:

$\mathbf{T}$	$\mathbf{t}^*$ at $u=1$	Y after intervention	flip?
$\emptyset$	—	1	no
$\{C\}$	0	0	yes
$\{D\}$	1	1	no
$\{C, D\}$	(0,1)	1	no

$\mathbf{T}$	$\mathbf{t}^*$ at $u=0$	Y after intervention	flip?
$\emptyset$	—	1	no
$\{C\}$	1	1	no
$\{D\}$	0	1	no
$\{C, D\}$	(1,0)	1	no

Cause $\mathbf{C} = \{P\}$, *intervention* $P \leftarrow 0$:

$\mathbf{T}$	$\mathbf{t}^*$ at $u=1$	Y after intervention	flip?
$\emptyset$	—	1	no
$\{C\}$	0	1	no
$\{D\}$	1	1	no
$\{C, D\}$	(0,1)	1	no

$\mathbf{T}$	$\mathbf{t}^*$ at $u=0$	Y after intervention	flip?
$\emptyset$	—	1	no

T	t* at $u=0$	Y after intervention	flip?
$\{C\}$	1	1	no
$\{D\}$	0	0	yes
$\{C, D\}$	(1,0)	1	no

The AC verdict per state matches Pearl. At $u = 1$, only $(\{X\}, \{C\})$ flips — the required witness $C = 0$ pins the cyanide path at its dormant value; X is the actual cause. At $u = 0$, only $(\{P\}, \{D\})$ flips — the witness $D = 0$ pins the dehydration path; P is the actual cause. So $U_{x,y} = \{u = 1\}$ and $U_{p,y} = \{u = 0\}$, exactly Pearl’s verdicts in his (10.15)–(10.16).

The PCI expectation. Under uniform Γ over the four **T**s and the uniform prior on u :

$$\mathbb{E}[\Phi \mid \mathbf{C}=\{X\}] = \frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot 0 = \frac{1}{8}, \quad \mathbb{E}[\Phi \mid \mathbf{C}=\{P\}] = \frac{1}{8}.$$

Equal responsibility, the expected reading for a symmetric preemption setup with a symmetric prior. Pearl Def. 10.3.5 says the same thing in its own units: $P(\text{caused}) = \frac{1}{2}$ for each cause.

Conditioning on the “no cyanide” forensic report (so $P(u = 1 \mid e) = 1$):

$$\mathbb{E}[\Phi \mid \mathbf{C}=\{X\}, e] = \frac{1}{4}, \quad \mathbb{E}[\Phi \mid \mathbf{C}=\{P\}, e] = 0;$$

Pearl Def. 10.3.5 gives 1 and 0. Both methods flip the verdict to “shooter caused death, poisoner did not”. The sign and ordering match in both regimes. The magnitudes do not — PCI’s $\frac{1}{4}$ is the Γ -fraction of witnesses passing the necessity test, while Pearl’s 1 is the posterior probability of a single binary claim. The two readings answer different questions, but they ride together on the ranking.

4. What the PCI expectation replaces in the AC machinery

The agreement in §3 is reassuring, but the more interesting observation is what changes underneath. **AC’s necessity clause is an existential:** *there exists* a witness set **T** and an alternative value c' such that the intervention flips Y . To verify the existential, one in principle enumerates the candidate $(\mathbf{T}, c')$ pairs — this is the source of AC’s combinatorial hardness. Pearl Def. 10.3.5 then averages the indicator of that already-enumerated predicate over noise states, inheriting the enumeration unchanged.

PCI collapses both moves into one expectation. Φ is averaged over the joint $(\mathbf{C}, \mathbf{T}, \mathbf{u})$ under $\Gamma \otimes P_{\mathbf{U}}$. The existential “does *some* witness work?” becomes the quantitative “*what fraction* of witnesses work?” — the same information AC compresses into one bit, but exposed as a graded score with sampling-based estimation.

In the desert traveler this is small change: four candidate witness sets, all trivially enumerable. The point is the scaling. In a model with n candidate witness variables, AC’s existential demands checking up to 2^n subsets per noise state, and Pearl’s posterior over the AC indicator inherits that cost per evaluation. PCI’s expectation can be estimated at a Monte Carlo budget the user controls, with variance independent of the problem’s combinatorial size. The desert-traveler agreement is the small- n end of a curve whose large- n end is the computational gap that motivates PCI in the first place.

That is the *operational* difference. The next section shows a difference that is not operational but *substantive*: a place where the two methods give qualitatively different answers about who is responsible.

5. A variant where sufficiency matters

The desert traveler is a clean preemption case: at any single noise state, one of the two enemies is uniquely responsible, so necessity and sufficiency coincide and PCI's sufficiency factor Y^s adds nothing to the ranking. To exhibit the second contribution PCI makes — a real sufficiency reading, separable from necessity — we introduce noise downstream of the poisoning so that the poisoner's mechanism becomes statistically unreliable while the shooter's stays deterministic.

The variant: weak poison. Same enemies, one tweak. The cyanide is a small dose that turns out fatal only with probability $\alpha = 0.1$ (independent exogenous noise ξ); $\alpha = 1$ recovers Pearl's original. Structural equations:

$$c = P(u' \vee X'), \quad V_C = c \cdot \xi, \quad D = X(u \vee P'), \quad Y = V_C \vee D.$$

Two factual scenarios under the same observation $X = 1, P = 1, Y = 1$. Forensic analysis identifies the mechanism after the fact:

- *Scenario A — cyanide killed*: $u = 0$ (drank early), $\xi = 1$ (this dose happened to be fatal).
- *Scenario B — dehydration killed*: $u = 1$ (drank late), ξ irrelevant.

Def. 10.3.5 makes the two scenarios look identical. Given the forensic information, the AC indicator is true with posterior certainty in each case: $P(\text{caused}(p, y \mid e_A)) = 1$ in A, $P(\text{caused}(x, y \mid e_B)) = 1$ in B. Each enemy is the actual cause of his victim, full probability, end of story.

PCI's necessity factor matches Pearl — the necessity component flips for P in A and for X in B, with the same witness analysis as §3. So far the two methods produce the same ranking.

PCI's sufficiency factor distinguishes the scenarios. The sufficiency leg reads off the configuration-level analogue of Pearl's Ch. 9 PS — *how reliably does this cause reinstate the outcome from a baseline world where both the cause and the outcome were absent?* Formally,

$$\text{PS}(\bullet) = P(Y_{\bullet=1} = 1 \mid \bullet = 0, Y = 0).$$

- *Poisoner*: condition on $P = 0, Y = 0$. Intervene $\text{do}(P = 1)$. The traveler is now poisoned but only dies if $\xi = 1$, so

$$\text{PS}(P) = \alpha = 0.1.$$

- *Shooter*: condition on $X = 0, Y = 0$. Intervene $\text{do}(X = 1)$. Two sub-cases collected over the conditioning posterior: if no poison was present ($P = 0$), the empty canteen kills via dehydration deterministically; if poison was present but $\xi = 0$, the shot turns out lethal with probability $P(u = 1) = 0.5$. Averaging gives

$$\text{PS}(X) \approx 0.76.$$

The two enemies' sufficiency probabilities differ by an order of magnitude. With a *ci* that combines necessity and sufficiency (the PNS-product *ci* of Section \ref{sec:causal_impact}, or any monotone composition of Y^s and Y^n), the two scenarios read:

Scenario	Def. 10.3.5	PCI sufficiency PS	PCI composite
A — poisoner caused	1 (certain)	0.1	low
B — shooter caused	1 (certain)	0.76	high

The intuitive payoff. A juror weighing the two deaths would treat the shooter in B as more clearly to blame than the poisoner in A. The shooter’s act was overwhelmingly likely to bring about the outcome; the poisoner’s was overwhelmingly likely *not* to — the poisoner needed both the right circumstances (early drinking) and a lucky cyanide outcome on top, neither of which were robust features of his action. PCI matches this intuition because it has a sufficiency factor to read off. Def. 10.3.5 does not, because the AC indicator at its core does not separate counterfactual reinstatement from counterfactual removal. (Pearl’s Ch. 9 has both PN and PS as separate quantities, but his Ch. 10 probability-of-actual-causation glues necessity-only AC verdicts to a posterior over $\mathbf{u}$ and drops PS on the way.)

Caveat on the kernel. The discrimination above requires (i) downstream noise — in a strictly deterministic SCM the sufficiency factor in the PNS-binary kernel collapses to factivity and the contrast disappears — and (ii) a *ci* that reads Y^s as a probability rather than as a binary indicator. The PNS-binary *ci* suffices; the contrast is more legible with the PNS-product or absolute-difference *ci* of Section \ref{sec:causal_impact}.

6. Other differences, more abstractly

The two worked examples have surfaced the load-bearing differences: PCI replaces AC’s existential with an expectation (§4), and PCI carries a separable sufficiency factor that Def. 10.3.5 lacks (§5). Three further differences are worth flagging but don’t need their own worked examples.

Definitional stability. The AC predicate Pearl plugs into Def. 10.3.5 has shifted three times in the literature — the original beam definition, Halpern–Pearl 2005, and the normality refinement of Halpern (2015) and Icard et al. (2017). Each shift moves the value of $P(\text{caused}(x, y \mid e))$ because the AC indicator inside changes. PCI is invariant to this patch-history because it embeds no AC predicate at all; where AC alignment is specifically wanted, Proposition \ref{prop:filter} recovers it through the expectation rather than through a definitional choice baked into the estimand. The desert-traveler agreement of §3 is genuine, but it is contingent on which AC variant one is reading; PCI’s score doesn’t shift if the literature shifts.

Breadth of aggregation. Def. 10.3.5 aggregates over one dimension: the noise $\mathbf{u}$, conditional on evidence e , for a claim $(X = x, Y = y)$ already fixed. PCI aggregates over four: the suspect set $\mathbf{C}$ and witness set $\mathbf{T}$ via Γ , the alternative value $\mathbf{c}'$ via Δ , and the noise via $P_{\mathbf{U}}$. This is what supports per-feature comparison — the same PCI expectation, evaluated over varying $\mathbf{C}$ inside a fixed $\mathbf{S}$, returns the relative responsibility of each candidate cause. Def. 10.3.5 cannot perform that comparison without a separate evaluation per claim, each inheriting the AC-predicate cost discussed in §4.

Continuous variables. The AC predicates inside Def. 10.3.5 are event-shaped: they read off $\text{AC}(x, y; M, \mathbf{u})$ for propositional x, y , and extensions to continuous variables require discretising the relevant event before the predicate applies. PCI’s *ci* function is continuous-native (the absolute-difference impact score of Section \ref{sec:causal_impact}), and the synthetic benchmark of Section \ref{sub:synthetic} exercises that continuity directly.

7. Summary

Pearl Def. 10.3.5 and PCI share the same explanatory target — probabilistic responsibility built on top of structural information — but instantiate it through different mechanisms. Def. 10.3.5 averages a binary AC indicator against a posterior over noise; PCI integrates a continuous necessity–sufficiency kernel against a joint distribution over suspect sets, witness sets, alternative values, and noise.

On clean preemption cases (§3) the two agree on ranking. They diverge in two substantive places. **The expectation replaces an existential** (§4): PCI’s Γ -weighted sum over witnesses surfaces the same information AC’s existential hides, but as a graded score whose estimation cost is independent of the combinatorial size of the witness search. **The sufficiency factor is separable in PCI but absent from Def. 10.3.5** (§5): when two causes differ in how reliably they reinstate the outcome — as the shooter and the poisoner do in the weak-poison variant — PCI separates them in a way that matches juror-style responsibility intuitions, while Def. 10.3.5 cannot. Three further differences (definitional stability, breadth of aggregation, continuous variables) reinforce the same conclusion: the two constructions answer related but distinct questions, and the question PCI answers is the one we need for ML-grade causal attribution.

Placement (decided in conversation)

- **§2 motivations — early flag (Proposal A0)**: short forward-reference paragraph between the section’s opening framing (`sections/sec2_motivations.tex:6`) and the `\subsection{Actual Causality}` heading (line 8). Job: plant a flag the first time the reader encounters §2, so the substantive discussion in §7 doesn’t come out of nowhere.
 - **§2 motivations — substantive paragraph (Proposal A)**: one paragraph between the PNS table and `\subsection{Path forward}` (inserted at `sections/sec2_motivations.tex:286`). Job: pre-empt “but Pearl already did this” before §3 opens.
 - **§7 Relation with Actual Causality (Proposal B)**: new subsection appended after `sections/sec7_actual_causality.tex:454`. Job: the technical contrast in detail.
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Proposal A0 — §2 early flag (LaTeX-ready)

Pearl’s own framework also has a probabilistic-actual-causation construction --- Definition~10.3.5 of `\citep[Ch.~10]{causalityPearl}`, which sums an actual-cause indicator against a posterior over noise states --- that is the closest neighbour to `\ourapproach{}` in the existing literature. We restrict attention in this section to the simpler PN/PS/PNS quantities of Chapter~9 that motivated it, and return to Definition~10.3.5 in Section~\ref{sub:pearl_actual_cause_prob} once the necessary machinery is on the page.

Comments / questions for A0

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Proposal A — §2 paragraph (LaTeX-ready)

Pearl himself addresses this gap in `\citep[Ch.~10]{causalityPearl}`: the `\emph{probability of actual causation}`, $P(\text{caused}(x,y \mid e)) = P(U_{xy} \cap U_e) / P(U_e)$ (Definition~10.3.5), where U_{xy} collects the noise states $\mathbf{u}$ in which a chosen actual-cause predicate holds. The diagnosis is the one we just made: PN/PS/PNS are global features of $Y_x(\mathbf{u})$ and miss the structural detail that distinguishes Alice's case from Bob's, so probabilistic responsibility has to be built on top of a scenario-specific predicate rather than on the marginal counterfactual alone. Pearl's construction takes a binary actual-cause predicate --- the causal-beam definition or the Halpern--Pearl variant `\citep{actualCausalityHalpern}` --- and averages its indicator against the posterior $P(\mathbf{u} \mid e)$. `\Ourapproach{}` shares the diagnosis but neither the binary indicator nor the AC predicate at its core: it integrates a continuous necessity--sufficiency kernel against a joint distribution over suspect sets, witness sets, alternative values, and noise. Section~`\ref{sec:relation_with_ac}` develops the contrast in detail; we flag here only that Definition~10.3.5 is `\ourapproach{}`'s closest neighbour in the literature, and that the two differ on each of the four axes that govern tractability and the kind of question that can be answered.

Comments / questions for A

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Proposal B — §7 subsection (LaTeX-ready)

`\subsection{Comparison with Pearl's Probability of Actual Causation}`
`\label{sub:pearl_actual_cause_prob}`

Pearl's Definition~10.3.5 in `\citep{causalityPearl}` proposes a probabilistic generalisation of actual causation that addresses the same diagnostic gap motivating `\ourapproach{}`: PN/PS/PNS are global features of the response function $Y_x(\mathbf{u})$ and miss scenario-specific structural information, so probabilistic responsibility ought to be built on top of a structural predicate. Concretely, for a chosen actual-causation predicate $\mathrm{AC}(x,y;M,\mathbf{u})$ --- the causal-beam definition of `\citep[Ch.~10]{causalityPearl}` in the first edition, the Halpern--Pearl 2005 AC1--AC3 in the second-edition postscript and in `\citep{actualCausalityHalpern}` --- Pearl defines

$$P(\text{caused}(x,y \mid e)) \stackrel{\text{def}}{=} \frac{P(U_{xy} \cap U_e)}{P(U_e)},$$

$$U_{xy} = \{\mathbf{u} : \mathrm{AC}(x,y;M,\mathbf{u})\},$$

the posterior expectation of the AC indicator under $P(\mathbf{u} \mid e)$. The same construction underwrites Chockler and Halpern's degree of blame `\citep{chocklerResponsibilityBlameStructuralModel2004}`, which substitutes a graded responsibility share inside the expectation. Because `\eqref{eq:pearl-paac}` is the closest

construction in the literature to `\ourapproach{}`'s expectation in Definition-`\ref{def:jointnecsuf}`, the differences are worth recording in detail.

`\medskip\noindent\textit{(i) Estimand shape.}`
`\eqref{eq:pearl-paac}` is a posterior expectation of the binary indicator $\mathbb{I}[\mathrm{AC}]$; its value is a probability that a fixed claim `` $X=x$ caused $Y=y$ '' is true under the model. `\ourapproach{}`'s $\mathbb{E}_{\Gamma \otimes \Delta \otimes P_{\mathbf{U}}}[\Phi]$ is the expectation of a continuous kernel that combines a sufficiency factor γ^s and a necessity factor γ^n with a Γ -weight; its value is a graded responsibility that ranks candidate causes against each other, rather than evaluating a sharp predicate. The two notions answer different questions: `\eqref{eq:pearl-paac}` answers ``how probable is the AC verdict?'' for a claim already on the table;  $\mathbb{E}[\Phi]$  answers ``how much of the outcome's causal weight does this variable carry?'' `\ourapproach{}` therefore returns interpretable comparisons across features --- the use case for the synthetic and AVM evaluations of Sections-`\ref{sub:synthetic}` and `\ref{sec:avm}` --- where `\eqref{eq:pearl-paac}` returns a single number per pre-specified claim.

`\medskip\noindent\textit{(ii) Tractability and definitional stability.}`
The cost of evaluating `\eqref{eq:pearl-paac}` is the cost of the AC predicate it embeds, and the structure-based AC family is intractable in general `\citep{eiterComplexityResultsStructurebased2002}`. `\ourapproach{}`'s kernel Φ embeds no AC oracle: evaluating it at a sampled $(\mathbf{C}, \mathbf{c}', \mathbf{T}, \mathbf{u})$ costs one twin-network counterfactual per factor, and the expectation is approximated by Monte Carlo over Γ , Δ , and $P_{\mathbf{U}}$. The corresponding stability point is independent: the AC predicate plugged into `\eqref{eq:pearl-paac}` shifted three times --- the original beam definition, Halpern--Pearl 2005, and the normality refinement `\citep{Halpern2015-HALGCA, icardNormalityActualCausal2017}` --- and each shift moves the value of $P(\text{caused}(x, y \mid e))$. `\ourapproach{}` is invariant to the AC patch-history because it embeds no AC predicate at all; where AC alignment is specifically wanted, Proposition-`\ref{prop:filter}` recovers it through the expectation rather than through a definitional choice baked into the estimand.

`\medskip\noindent\textit{(iii) Breadth of aggregation.}`
`\eqref{eq:pearl-paac}` aggregates over a single dimension: the noise $\mathbf{u}$, conditional on evidence e , for a claim $(X=x, Y=y)$ already fixed. `\ourapproach{}` aggregates over four: the suspect set $\mathbf{C}$ and witness set $\mathbf{T}$ via Γ , the alternative value $\mathbf{c}'$ via Δ , and the noise $\mathbf{u}$ via $P_{\mathbf{U}}$. This is what supports per-feature comparison: Γ exposes the choice of which set to ask about as a first-class component, so the same expectation evaluated over varying $\mathbf{C}$ inside a fixed $\mathbf{S}$ returns the relative responsibility of each candidate. `\eqref{eq:pearl-paac}` cannot perform this comparison without a separate evaluation per claim, each inheriting (ii)'s cost.

`\medskip\noindent\textit{(iv) Continuous variables and symmetric sufficiency.}`
The AC predicates inside `\eqref{eq:pearl-paac}` are event-shaped: they read off $\mathrm{AC}(x, y; M, \mathbf{u})$ for propositional x, y , with extensions to continuous

variables discretising the relevant event. `\Ourapproach{}`'s `$ci$` function is continuous-native (see the absolute-difference impact score in Section~\ref{sec:causal_impact}), and the synthetic benchmark of Section~\ref{sub:synthetic} exercises that continuity directly. Sufficiency enters `\eqref{eq:pearl-paac}` only as part of the AC predicate's sustenance leg, which is a structural-contingency check rather than a counterfactual-reinstatement probability `\`a la` Pearl PS; `\ourapproach{}` keeps the necessity factor `Y^n` and the sufficiency factor `Y^s` separate in the kernel, both indexed by `$(\mathbf{C},\mathbf{c}',\mathbf{T},\mathbf{u})$`, and the synthetic benchmark shows that the two factors discriminate causal archetypes that PNS-only and AC-only scores conflate.

`\medskip`

Taken together, Pearl's Definition~10.3.5 and `\ourapproach{}` share an explanatory target --- probabilistic responsibility built on top of structural information --- but instantiate it through different mechanisms: an AC-indicator expectation against `$P(\mathbf{u} \mid e)$` in the first case, a continuous necessity--sufficiency kernel against `$\Gamma \otimes \Delta \otimes P_{\mathbf{U}}$` in the second. The first is the natural construction if a sharp AC predicate is given and a posterior over `$\mathbf{u}$` is the only object one wants to integrate against; the second is the natural construction if responsibility is to scale to ML-grade models and to support cross-feature comparison.

Comments / questions for B

(leave inline comments below this line)

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Notes / flags

- **Citations used** (all already in `references.bib`, no new entries needed): `causalityPearl`, `actualCausalityHalpern`, `chocklerResponsibilityBlameStructuralModel2004`, `eiterComplexityResultsStructurebased2002`, `Halpern2015-HALGCA`, `icardNormalityActualCausal2017`
- **Soft complexity claim, deliberate.** I wrote “structure-based AC family is intractable in general” rather than naming a complexity class. Eiter–Lukasiewicz 2002 proves Σ_2^P -completeness for HP 2001; HP 2005 and the normality-refined variants are in the same neighbourhood but the cleanest defensible claim is intractability. Promote to the sharper Σ_2^P claim?
- **Length.** Subsection B is ~75 lines of LaTeX, ~2/3 page. Cut path if needed: collapse (i)+(iii) and (ii)+(iv) into two paragraphs instead of four.
- **Chockler–Halpern degree of blame.** Cited once in B.0 but really is the existing literature’s “graded responsibility” notion. Should it get its own paragraph in §11 related-work, or a footnote in §7? Currently a one-line mention is light.
- **Forward references resolved.** A: `sec:relation_with_ac`. B: `def:jointnecsuf`, `prop:filter`, `sub:synthetic`, `sec:avm`, `sec:causal_impact`. All exist.

- **Decisions pending.** (1) approve A, B, both, or revisions; (2) promote complexity claim?; (3) Chockler–Halpern follow-up; (4) any preferred trim of B.